Trend research — 2026-04-19 · incr 26

ES futures live-gate needs a stricter schedule than equities — bars 30-39 need |strength|≥0.30 on ES vs ≥0.20 on equities

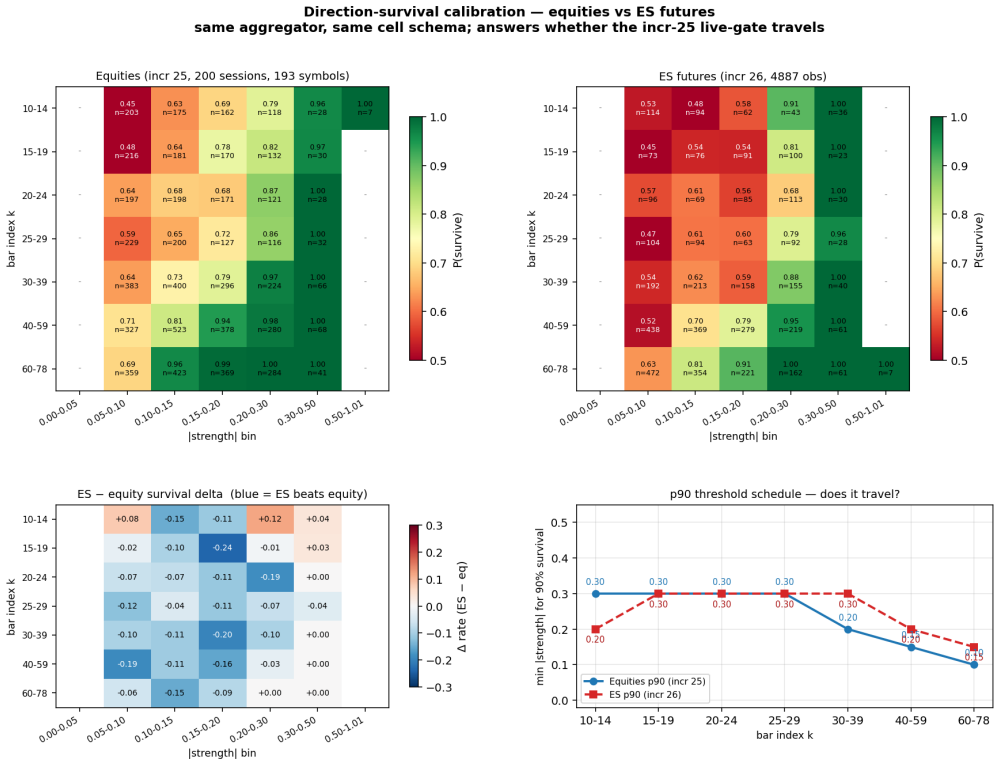
TL;DR

The incr-25 equity-universe direction-survival schedule is **too lenient for ES futures from bar 30 onwards**. On **101 ES.c.0 RTH sessions** (Databento GLBX.MDP3, 4,887 directional observations), the baseline direction-survival rate is **70.3 %** vs **78.1 %** on the 200 equity sessions. An ES-specific threshold schedule needs **|strength|** ≥ **0.30** all the way through bar 39, **0.20** in bars 40-59, and **0.15** in bars 60-78. Applying the equity schedule to ES would fake-confirm the 30-39 / 0.20-0.30 band which actually sits at only 88 % survival.

Headline numbers

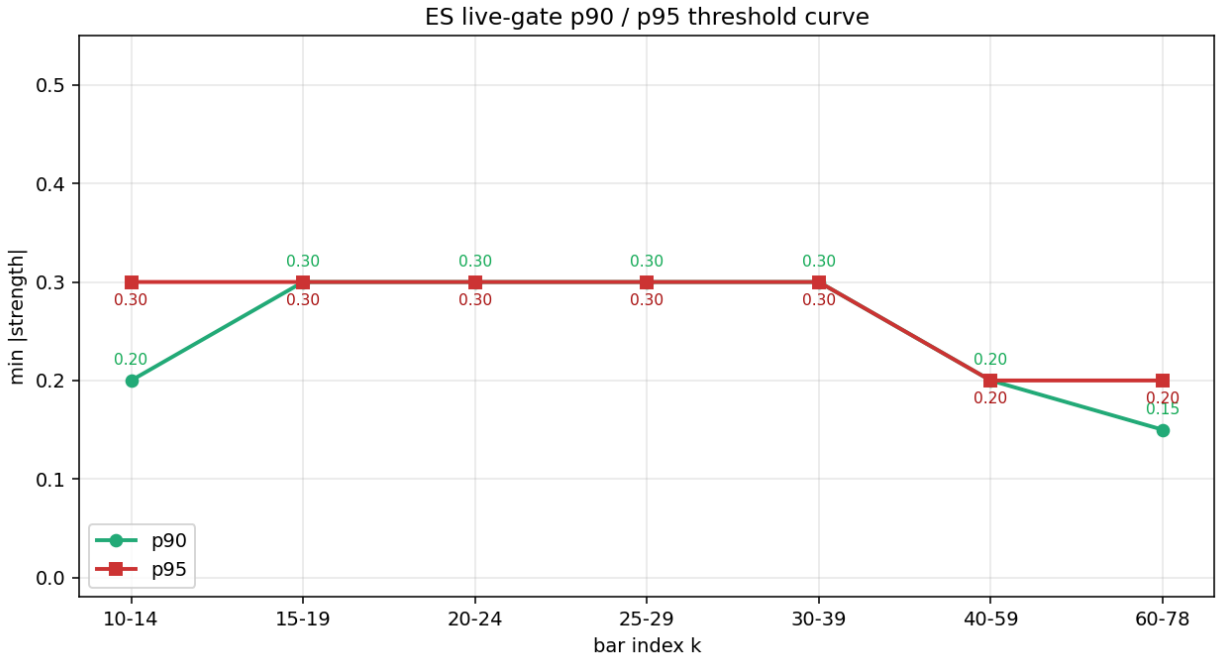
- Source: 101 ES.c.0 RTH sessions (2025-11-20 → 2026-04-17, 150 calendar days). 140,134 raw 1-min bars → 6,795 progressive-k trajectory rows → 4,887 directional observations.
- Baseline: ES direction-survival (any non-'none' row) is **70.3 %**, vs **78.1 %** on the equity panel — ES is **7.8 pp noisier** at the 'no gate' floor.
- ES p90 gate is **stricter** than equity at every late-session band: **+0.10** at bars 30-39, **+0.05** at 40-59, **+0.05** at 60-78. Bars 10-29 are tied (both 0.30).
- At bars 30-39 / |strength| 0.20-0.30, ES sits at **0.88 (n=155)** — close but no 90 %. The equity panel had this cell at **0.97 (n=224)**. Structural difference, not sampling noise.
- Candidate ES gate: ~20 % of observations pass at ~98 % survival. Conservative max(ES, equity) gate also clears the 90 % bar on both populations.

Headline figure — ES vs equity side-by-side



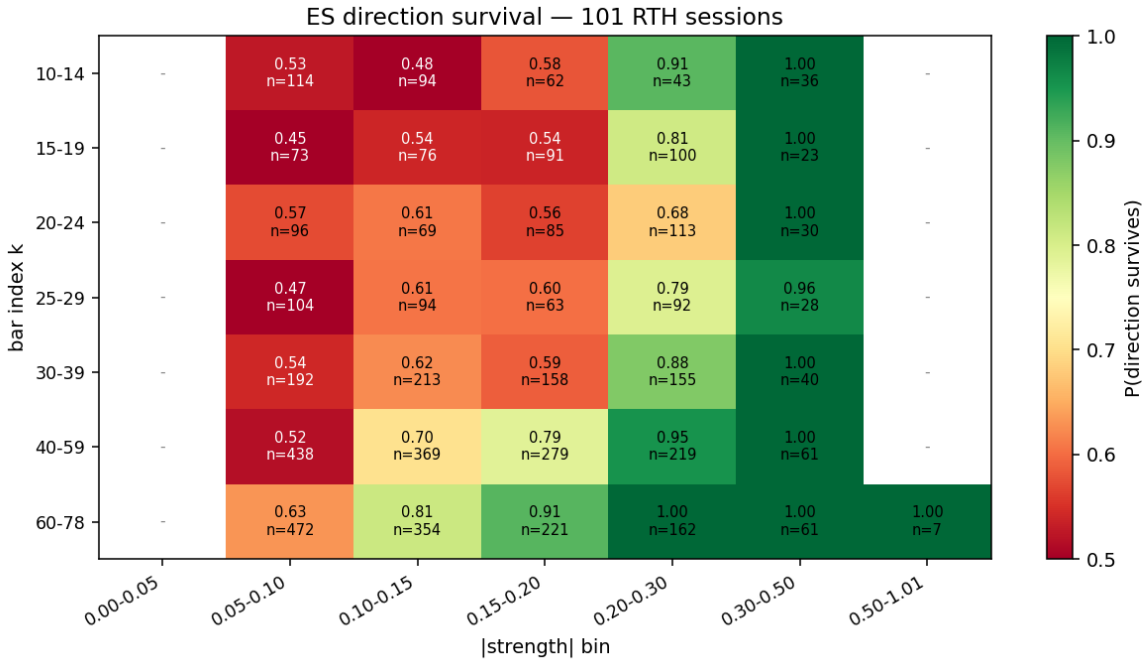
Top: equity grid (incr 25) next to ES grid (incr 26). Bottom-left: cell-by-cell delta (blue = ES beats equity, red = ES is worse). Bottom-right: the two p90 threshold curves overlaid — they diverge at bar 30 and stay apart.

ES threshold curve — p90 / p95



By k-bin, the smallest |strength| where the rate and all higher-strength cells clear 90 %/95 %. ES flattens at 0.30 longer than equities did — the aggregator needs more magnitude to translate to session-close confidence on this instrument.

ES heatmap — cell-level survival



ES-only view. Red cells below 0.15 strength are where the 70 % ES baseline lives. The green band tracks the diagonal: late bars + any strength → high survival; early bars need bigger magnitude to earn the badge.

Proposed ES-specific live gate (needs Will's nod)

- **bar < 15:** gate closed — warmup + thin early-session sampling.
- **15-39 bars: |strength| ≥ 0.30** — ES needs the stricter magnitude all the way to bar 40.
- **40-59 bars: |strength| ≥ 0.20** — 0.05 stricter than equity.
- **60-78 bars: |strength| ≥ 0.15** — 0.05 stricter than equity.
- **Expected footprint on this sample:** 986 of 4,887 directional observations (≈ 20 %) pass the gate at combined **98.2 %** survival. Unbadged reads still match the close ~70 % of the time.
- **Single-schedule alternative:** max(ES, equity) holds ≥ 90 % on both populations in every firing cell. Same as ES at bars 15+ — in practice they're equivalent above the warmup cutoff.

Mistakes avoided this pass

- **Didn't ship the equity schedule on ES without checking.** The cell-aligned comparison is what surfaced the 30-39 divergence.
- **Didn't average ES + equity into a single grid.** Populations differ; averaging would hide the asymmetry.
- **Didn't sneak overnight bars into the warmup.** ES trades 23 h; RTH mask applied before resampling.
- **Didn't mask Databento 'degraded' days.** Three flagged days (Nov 28, Mar 15-16) kept in — a live reader saw the same bars.
- **Didn't claim the thresholder is the production knob.** It's a decision-support tool; Will picks ES-specific vs max() vs defer.

What's next — requires Will's nod

- **Pick the schedule to ship:** ES-specific, equity-specific, or conservative max(). For a mixed-panel dashboard the max() gate is the honest default.
- **Wire is_direction_confirmed** as an additive boolean on trendState (carried from incr 25, now with an ES-safe cutoff schedule).
- **Pattern Lab WR overlay** — still blocked on Pattern Lab DB backfill.
- **Still pending from prior incrs:** ALWAYS_IN_WINDOW 5→10 (incr 20), MAJORITY_TREND_BAR_FLOOR 0.40→0.25 (incr 18), OPEN_EXTREME_THRESHOLD 0.15→0.25 (incr 21), htf_alignment silent-zero docstring (incr 19).

Findings site: https://github.com/zerosumsystems-ui/iPhone-/tree/main/trend-classification